Karen Xiao

Incoming CS PhD Student at UC San Diego

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EDUCATION

University of California, San Diego | La Jolla, CA
Ph.D. Computer Science | *2026-2030*
Advisors: Leo Porter, Gerald Soosairaj, Bill Griswold

Wellesley College | Wellesley, MA
B.A. Computer Science and Mathematics, *Summa Cum Laude* | *2023-2026*
Departmental Honors in Computer Science (Honors Thesis)

University of Maryland, College Park | College Park, MD
Honors College, Computer Science, GPA: 3.97/4.0 | *2022 –2023*
Gemstone Research Program

RELEVANT COURSES

Human Computer Interaction; Natural Language Processing; Fundamental Algorithms; Deep Learning; Introduction to Machine Learning; Regression Analysis & Statistical Models; Theory of Computation; Computer Systems; Data Structures; Combinatorics & Graph Theory; Advanced Linear Algebra; Abstract Algebra; Multivariable Calculus; Real Analysis

RESEARCH EXPERIENCE

Senior Honors Thesis | Wellesley College
Advisor: Brian Brubach | *August 2025 – present*

- Completed a senior honors thesis that proposes methods to measure the diversity of evaluation in classes by treating different types of assignments as alternative ways of ranking students

NSF REU Intern | Carnegie Mellon University
Human Computer Interaction Institute | *May 2025 – September 2025*

- Worked with Conrad Borchers and Dr. Vincent Aleven in the Creating Adaptive Tutoring Systems (CATS) Lab
- Applied statistical methods to analyze the impact of data-driven redesign on intelligent tutoring systems
- Identified new insights into what constitutes a productive level of difficulty for data-optimized practice

Research Assistant | Wellesley College
Institute for Mathematics and Democracy | *December 2024 – September 2025*

- Simulated various ranked choice election methods on datasets containing ballots for thousands of real elections
- Conducted empirical comparison of various ranked choice voting methods using Python and VoteKit
- This research is believed to be the largest study of ranked choice elections ever performed

Undergraduate Student Researcher | Wellesley College
CrowdNets Lab | *August 2024 - May 2025*

- Defined and evaluated a graph-based spatio-temporal POI ranking model that outperforms baseline ML models
- Ran simulations and designed dynamic visualizations on large-scale real-world Uber data to evaluate performance
- Won 2nd place in the Undergraduate Student Research Competition at ACM SIGSPATIAL 2025 (Minneapolis, MN)

Undergraduate Student Researcher | MIT
MIT Urban Metabolism Group | *February 2024 - May 2024*

- Collected and analyzed data regarding climate risks and policies in various metropolitan clusters
- Converted a policy portfolio into a quantifiable matrix for ML-driven insights

PUBLICATIONS

Liu, Q., Borchers, C., Xia, M., **Xiao, K.,** Carvalho, P., Koedinger, K., & Aleven, V. (2026). Evaluating a Process for the Data-driven Redesign of Educational Technology: Improved Learning Processes. In *The 27th International Conference on AI in Education (AIED '26). https://doi.org/10.1007/978-3-032-29760-0_12*

Xiao, K., Irisumi, M., & Bassem C. (2025). A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms. In *The 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25). https://doi.org/10.1145/3748636.3766535* **[SRC 2nd Place Winner]**

WORK EXPERIENCE

Financial Data Science Internship — Quantbot Technologies LP
Data Science Intern — *May 2024 - August 2024*

- Optimized data capture and validation process using DuckDB, Pandas, and Apache Iceberg
- Assessed, enriched, and normalized company outlook and projection data covering all global markets
- Researched new database query tools that the company can implement to optimize data capture toolkits

Financial Data Science Internship — Quantbot Technologies LP
Data Science Intern — *May 2023 - August 2023*

- Validated and refined raw Equities and Futures data using Python Pandas and NumPy on AWS
- Collaborated with quantitative researchers, traders, and external data vendors
- Optimized existing data processing toolkits using SQLite and Pandas

LEADERSHIP EXPERIENCE

Undergraduate Teaching Assistant — Wellesley College
Wellesley College Computer Science Department — *January 2024 - present*

- Lead weekly office hours, offer 1-on-1 tutoring, and grade assignments for CS 231 Design & Analysis of Algorithms over the course of 4 semesters

Captain — Wellesley College
Wellesley College Club Squash Team — *August 2025 - present*

- Lead and organize a team of 15 players by coordinating practice and matches, mentoring new players, and fostering a positive team environment

Networking Chair — Wellesley College
Wellesley Computer Science Club — *October 2024 - present*

- Manage alumni and recruiting connections to organize and communicate important recruiting, networking, and volunteering events to underrepresented students interested in CS opportunities

AWARDS, FELLOWSHIPS, & GRANTS

Conference Funding

- (2025) **SIGSPATIAL Travel Award (SIGSPATIAL '26) - $1,000,** National Science Foundation
- (2025) **Science Center Student Conference Travel Grant - $300,** Wellesley College

Awards

- (2026) **Excellence in Teaching Award in Computer Science,** Wellesley College
- (2026) **NSF GRFP Honorable Mention**, National Science Foundation
- (2025) **Undergraduate SRC 2nd Place Winner**, ACM SIGSPATIAL '26